

Aurora Bank Analysis

Technical Report

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Abstract

This technical report documents the end-to-end Business Intelligence developed for Aurora Bank to identify high-risk borrowers and support data-driven decision-making. The solution follows the Business Intelligence value chain, beginning with identifying business problems to decision-making to create business value. In this report, analysis and business intelligence processes will be focused mainly on.

Introduction

Aurora Bank operates in a financial environment where responsible lending and risk monitoring are essential. The 2008 financial crisis demonstrated that relying on limited indicators, such as credit scores alone, can lead to poor lending decision and increased default risk. Modern bank therefore needs a more holistic approach to borrower evaluation, combining demographic, financial, behavioral, and transactional indicators.

The purpose of this technical report is to document the development of end-to-end Business Intelligence solution for Aurora Bank. The technical architecture of the solution followed a clear BI pipeline:

**Raw Data → Power Query → Cleaned Tables → Power BI Data Model → DAX Measures → Dashboard
→ Prediction Model → Business Insights**

The raw data files were first imported into Power BI. Power Query was used for cleaning, type conversion, formatting, filtering, and transformation. After the data was prepared, the cleaned tables were loaded into the Power BI data model. Relationships were then created between the fact table and dimension tables. DAX measures and calculated columns were added to support KPI calculations, risk analysis, and dashboard visuals. Finally, a prediction model was added to provide in-depth insight.

Analysis

Exploratory Data Analysis

Data Sources

The dataset was obtained from a Data Analytics group on Facebook. On this site, people share the datasets that they used to work on and enhance their professional profile. After researching different sources and

types of data, we finalize this dataset from Aurora Bank to work with. There are four structures data files and each file represented a different business area and was used as either a fact table or dimension table in the final model. These tables were stored under excel files without any initial connection. A clear description of dataset was provided in (Appendix A).

Data Preparation

Before working on this project, we were working on two different projects, but both were eliminated at this stage. The first project was about fast fashion, main focus was Zara industrial position, the dataset was found on Kaggle and was a synthetic one. We cleaned the data and used Python with Pandas to conduct the analysis but when we came to the Exploratory Data Analysis (EDA), the visualization was meaningless which proved that the data itself has no value. The same experience happens for the second project which is Supply Chain (Appendix B).

For this project, Data Cleaning was conducted mainly in Power Query before creating the analytical model. Our group's strategy is to use Power BI as the main tool to interpret this project, so most of the data cleaning processes were done here. Excel and Python acted as complementary tools for cross-checking when a field was unclear in Power BI.

Financial columns were reviewed and converted into numeric format where necessary. Some values were originally stored as text, which would prevent correct aggregation and calculation. Converting these fields into numeric data types allowed measures such as total spending, yearly debt payment, yearly income, and debt-to-income ratio to be calculated correctly. Transaction date was also standardized. A clean transaction date field was created so the transaction table could be connected to a separate date dimension. Geographic fields such as latitude and longitude were reviewed to identify invalid or implausible values. Incorrect coordinates were converted to null values to reduce the risk of misleading map-based visualizations. Transaction error fields and merchant descriptors were also fixed to standardize some fields such as failed transactions or insufficient balance errors. Cleaning these fields ensured that transaction failures were counted consistently.

After cleaning, several new variables and dimensions were created to support analysis. These variables converted raw data into more meaningful analytical categories.

A date dimension was created to support structured time intelligence. A merchant category codes (MCCs) dimensions were also added for a clear demonstration of the involved business type (CFA et al., 2026). Customer risk variables were added such as yearly debt payment, and with this value we can obtain debt-to-income ratio (Appendix C). This measure was then grouped into DTI buckets for low, moderate, high and very high financial pressure. Credit score values were also grouped into categories such as poor, fair, good, and excellent. Additional segmentation variables were created, including age group, income band, and recommended action group. These categories made the dashboard easier to interpret because users could compare groups rather than individual numeric values.

Data Model

The solution used a star-schema-inspired model. The Fact_Transaction table was placed at the center of the model because it contained individual transaction records. The supporting dimension tables were connected around it. A star-schema model is provided in (Appendix D).

Single-direction relationships were used to reduce ambiguity in filtering behavior. The dashboard included multiple slicers and cross-filtering visuals so single-directional relationships would help to enhance the interconnection between dimensions. A clean relationship structure helped ensure that filters produced reliable results and that measures were calculated consistently.

The star-schema-inspired design also improved usability. Business users could filter transaction data by customer group, card type, merchant category, and time period without needing to understand the raw data structure.

Structure Data Analysis

After the Exploratory Data Analysis stage, Structured Data Analysis was performed to examine the dataset in a more systematic and business-oriented way. In this project, Structured Data Analysis was divided into two main types: trend analysis, mathematical analysis (AnalytixLabs, 2022). Each type supported a different part of the Business Intelligence value chain and helped transform prepared data into decision-support insights.

Trend Analysis

Trend analysis was used to identify patterns and changes in customer behavior over time and across different segments. In the Aurora Bank solution, trend analysis was applied to transaction volume, spending behavior, and transaction failure patterns. The visualization helps to identify periods with higher spending activity, changes in customer transaction behavior, and possible increases in failed transactions.

Trend analysis was also used to compare customer groups. For example, the dashboard allowed managers to understand whether certain customer segments consistently had higher spending, more transaction errors, or stronger signs of financial pressure.

From a technical perspective, trend analysis was supported by the Power BI date table, time-based filters, line charts, bar charts. These elements allowed users to interactively explore how customer behavior changed over time and across segments.

Mathematical Analysis

Mathematical analysis was used to calculate key performance indicators and risk-related measures. This stage converted raw financial and transaction data into measurable indicators that could be used for comparison and decision-making.

The most important mathematical calculation in the project was the debt-to-income ratio. The result was used to estimate the level of financial pressure on each customer. Customers with a higher DTI ratio were considered more financially constrained because a larger share of their income was already committed to debt repayment. Other mathematical measures included total spending, average transaction amount, total debt, transaction error count, and insufficient balance error count. These calculations helped quantify both financial behavior and possible distress signals.

Technically, this stage was mainly implemented using calculated columns, Power Query transformations, and DAX measures in Power BI. These calculations formed the foundation for dashboard KPIs, slicers, and risk-based visuals.

Business Intelligence

Explanation

A full interpretation has been performed in Digital Business Analysis.

Prediction Model

Standard credit scores often lack nuance because they mainly reflect a customer's past repayment behavior, but they do not fully show their current financial pressure. To create a more complete risk assessment, Aurora Bank should adopt a three-pillar Risk Categorization Framework that combines Debt-to-Income ratio (DTI), Credit Utilization ratio, and Credit Score. By using these three indicators together, the bank can better understand customer financial health and identify borrowers who may be at higher risk.

The first pillar is the Debt-to-Income ratio, which compares a customer's yearly debt payments to their gross yearly income. This ratio shows how much of a customer's income is already committed to debt repayment. A high DTI means the customer has less remaining income for daily expenses, savings, or unexpected costs. In this framework, 43% is used as the DTI threshold because it was the original cap established by the Consumer Financial Protection Bureau's Qualified Mortgage rule. Customers with a DTI of 43% or higher are considered more financially stretched and adding a mortgage payment could put you in a tight financial spot.

The second pillar is the Credit Utilization ratio, which measures how much of a customer's available credit is currently being used. It is calculated by dividing total debt by total credit limit. Lenders and scoring models like FICO and VantageScore use this ratio to assess financial responsibility. A high utilization ratio suggests that a customer is overly dependent on credit and has limited borrowing flexibility. Aurora Bank uses 50% as the threshold for credit utilization. Customers above this level are considered higher risk because they are using a large share of their available credit (Centier Bank, 2024).

The final pillar is the Credit Score, which remains an important indicator of financial reliability. Credit scores help lenders estimate how likely a borrower is to repay debts on time. However, credit scores should not be used alone. In this framework, Aurora Bank uses 579 as the credit score threshold. Customers with a

score of 579 or lower are classified as higher risk because they fall within the poor credit range and have a greater likelihood of delinquency (Louis, 2024).

Based on these three pillars, each customer is assigned a composite risk score correspondently: Low, Moderate, Elevated or High Risk (Appendix E). The total score ranges from 0 to 4 based on the following model:

- Debt-to-Income (DTI) ratio threshold: 43% or higher.
- Credit Utilization ratio threshold: 50% or higher.
- Credit Score threshold: 579 or lower.

Based on these criteria, each customer was assigned a composite Risk Score ranging from 0 to 3. A point is added to the score for every threshold the borrower exceeds, allowing us to rank their level of financial risk:

- Score 0 (Low Risk) represents customers who exceed no threshold.
- Score 1 (Moderate Risk) represents customers who exceed exactly one threshold.
- Score 3 (Elevated Risk) represents customers who exceed exactly two thresholds.
- Score 4 (High Risk) represents customers who exceed all three thresholds.

Conclusion

This technical report documented the end-to-end Business Intelligence value chain developed for Aurora Bank. The solution transformed raw banking data into a structured analytical model using Power Query, Power BI relationships, calculated fields, dashboard visualizations and predicting model.

The final dashboard supports risk-based decision-making by combining debt-to-income ratio, credit quality, and transaction-based financial distress indicators. This provides a more complete view of borrower risk than relying on credit scores alone. Overall, the BI solution demonstrates how Aurora Bank can use data modelling, analytical measures, and visualization to convert raw operational data into actionable business intelligence.

Reference

AnalytixLabs. (2022, July 15). *Business Intelligence Value Chain—The Process of Powerful Business*

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Centier Bank. (2024). *How Does Credit Utilization Affect Your Credit Score?*

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CFA, T. B. F. B. T. J. B. is a, investing, C. with more than 20 years of experience in various areas including,

Management, I. P., finance, accounting, Investment, P., Advice, F. P., Insurance, D. of E. M. A. L., &

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Appendix

Appendix A. Datasets

Table	Description	Role in BI Model
Dim_client (users_data)	Contains customer demographic and financial information.	Customer dimension
Dim_card (cards_data)	Contains card ownership and card type information	Card dimension
Dim_mcc (mcc_codes)	Contains merchant category code descriptions	Merchant dimension
Fact_transaction (transactions_data)	Contains detailed transaction-level records	Central fact table

Appendix B. Prior project

The first one was Zara fast fashion situation. The link of dataset is attached below. This is a synthetic dataset; we used Python with Pandas to evaluate the dataset to see if it was feasible to be used for analysis. It turned out that the dataset was unable to draw any insight since the information in it was redundant and unhelpful.

<https://www.kaggle.com/datasets/khushikyad001/the-true-cost-of-fast-fashion-impact>

The second project was Supply Chain; we were aiming to identify the best route for transportation and find a way to optimize the operation. Our biggest mistake was not to invest enough researching in terms of business problems so when we finally sat down to discuss about that, the business problem was not big impactful enough to tackle. It meant that if we insisted on pursuing this project, it will not lead to any valuable insight. The link of the dataset is attached below.

<https://www.kaggle.com/datasets/harshbhatt02/fast-fashion-supply-chain?select=Warehouse+Shipping+Costs.csv>

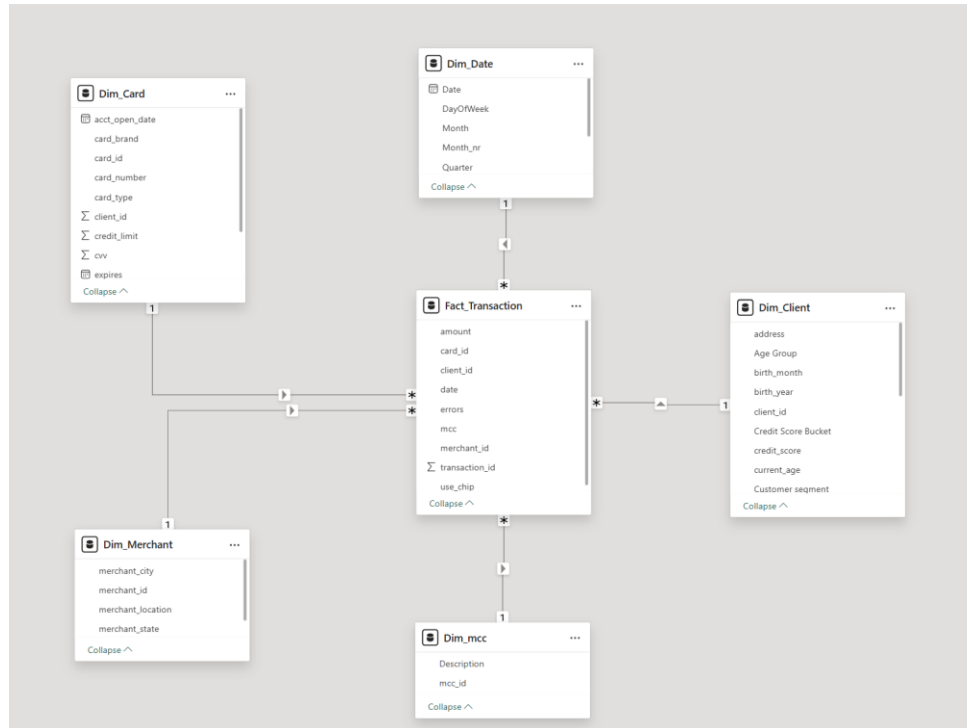
Appendix C. Debt-to-income ratio (DTI)

The rate is 4% and payment time zone is 5 years. Then we have the ratio is 0.225.

$$\text{Yearly debt payment} = \text{total debt} * 0.225$$

$$DTI = \frac{\text{yearly debt payment}}{\text{yearly income}}$$

Appendix D. Star-schema Model



Appendix E. Risk Categorization Framework

Based on three mentioned pillars, each customer is assigned a composite risk score. One point is added for each threshold the customer exceeds:

Risk Indicator	Threshold	Risk Point
Debt-to-Income Ratio	43% or higher	1
Credit Utilization Ratio	50% or higher	1
Credit Score	579 or lower	1

The total score ranges from **0 to 3**, and customers are categorized as follows:

Composite Risk Score	Risk Category	Interpretation
0	Low Risk	Customer does not exceed any risk threshold
1	Moderate Risk	Customer exceeds one risk threshold
2	Elevated Risk	Customer exceeds two risk thresholds
3	High Risk	Customer exceeds all three risk thresholds